

Introduction to Machine Learning

Lecture 9: Support Vector Classifier

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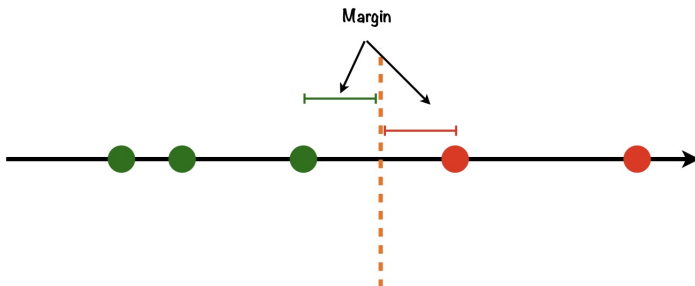
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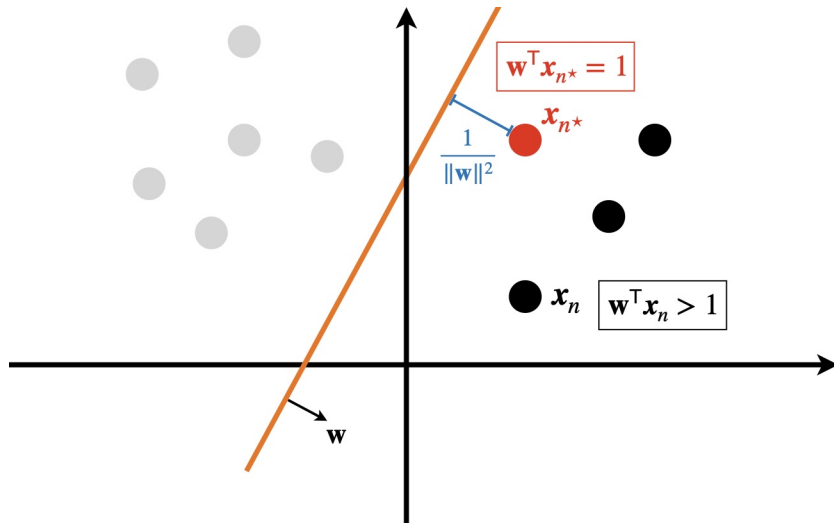
Quick Recap: *Classification with Confidence*

Classifying with Maximal Margin

We try to find linear classifier with maximal margin to *support vectors*



Quick Recap: Support Vector Classifier



Quick Recap: SVC Formulation

Training with maximal margin then looks like

$$\max_{\mathbf{w}} \frac{1}{\|\mathbf{w}\|} \quad \text{subject to } v_n \mathbf{w}^T \mathbf{x}_n \geq 1 \text{ for all } n$$

which can be alternatively written as

$$\text{maximum margin} \rightarrow \boxed{\min_{\mathbf{w}} \|\mathbf{w}\|^2} \quad \text{subject to } \boxed{v_n \mathbf{w}^T \mathbf{x}_n \geq 1} \leftarrow \text{no error}$$

Today's Agenda: *Support Vector Classifier*

Today, we find the solution to classification with maximal margin, i.e.

Support Vector Classifier

In this way, we discuss the following topics

- *We find out what the support vectors are*
- *We understand how SVC applies cross-validation*
- *We understand how to handle outliers and enable misclassification*
- *This motivates us to think about nonlinear patterns too*

Back to SVC: Dual Problem

In SVC, we want to solve the following problem

$$\min_{\mathbf{w}} \frac{1}{2} \|\mathbf{w}\|^2 \quad \text{subject to } v_n \mathbf{w}^\top \mathbf{x}_n \geq 1 \text{ for all } n$$

Maybe we can write it in the standard form as

$$\min_{\mathbf{w}} \frac{1}{2} \|\mathbf{w}\|^2 \quad \text{subject to } 1 - v_n \mathbf{w}^\top \mathbf{x}_n \leq 0 \text{ for all } n$$

We can see that **Slater's conditions** hold: *we can solve the dual problem*

$$\ell(\mathbf{w}, \boldsymbol{\lambda}) = \frac{1}{2} \|\mathbf{w}\|^2 + \sum_{n=1}^N \lambda_n (1 - v_n \mathbf{w}^\top \mathbf{x}_n)$$

Solving Dual Problem: *Stationary Points*

First, we find the stationary points

$$\begin{aligned}\nabla \ell(\mathbf{w}^*, \boldsymbol{\lambda}^*) = \mathbf{0} &\rightsquigarrow \mathbf{w}^* - \sum_{n=1}^N \lambda_n^* v_n \mathbf{x}_n = \mathbf{0} \\ &\rightsquigarrow \mathbf{w}^* = \sum_{n=1}^N \lambda_n^* v_n \mathbf{x}_n\end{aligned}$$

The optimal model is a weighted average of data-points

Solving Dual Problem: *Find Dual Optimal*

We could find the dual objective by replacing $\mathbf{w}^$*

$$\ell(\mathbf{w}^*, \boldsymbol{\lambda}) = \sum_{n=1}^N \lambda_n - \sum_{n=1}^N \sum_{m=1}^N \lambda_n \lambda_m v_n v_m \mathbf{x}_n^T \mathbf{x}_m$$

and maximize it, i.e.,

$$\max_{\boldsymbol{\lambda} \geq \mathbf{0}} \sum_{n=1}^N \lambda_n - \sum_{n=1}^N \sum_{m=1}^N \lambda_n \lambda_m v_n v_m \mathbf{x}_n^T \mathbf{x}_m$$

Or alternatively find feasible solutions to the complementary slackness

Key Observation

*Optimal model is characterized only via **cross-correlations***

Solving Dual Problem: Complementary Slackness

We can find λ_n^* also via complementary slackness: at all n we need to see

$$\lambda_n^* \left(1 - v_n \mathbf{w}^{*\top} \mathbf{x}_n \right) = 0$$

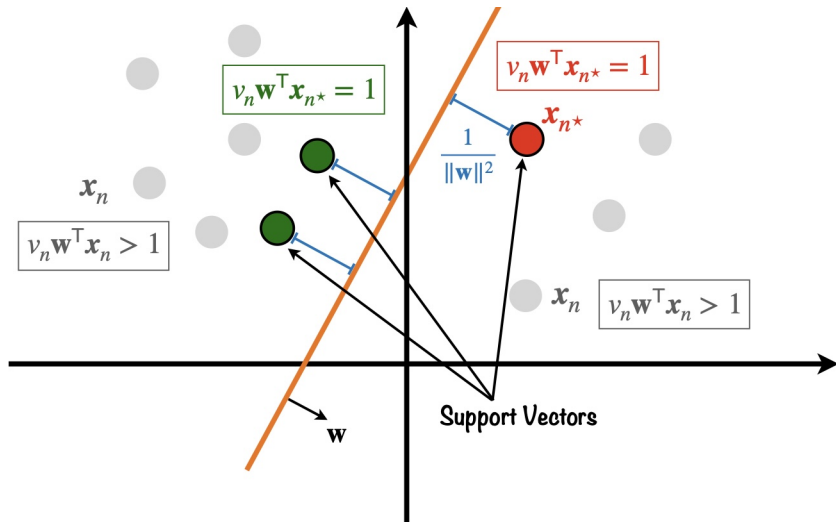
It describes system of equations that is solved uniquely via feasibility constraints

But, without solving it, we can say that

- If the constraint is active, i.e., $v_n \mathbf{w}^{*\top} \mathbf{x}_n = 1$; then, $\lambda_n > 0$
- If the constraint is inactive, i.e., $v_n \mathbf{w}^{*\top} \mathbf{x}_n > 1$; then, $\lambda_n = 0$

? Which constraints are active in SVC?!

Active Samples in Solution of SVC



Complementary Slackness: Only Support Vectors Matter

We can hence write the solution of SVC as

$$\begin{aligned}\mathbf{w}^* &= \sum_{n=1}^N \underbrace{\lambda_n^*}_{\neq 0 \text{ at Support Vectors}} v_n \mathbf{x}_n \\ &= \sum_{n \in \mathcal{S}} \lambda_n^* v_n \mathbf{x}_n\end{aligned}$$

This is why we call it **support vector** classifier

Conclusion

To find SVC, we first compute all **cross-correlations** $\mathbf{x}_m^T \mathbf{x}_n$

- Using **cross-correlations** we can find λ_n^*
 ↳ They are non-zero only of support vectors
- We set $\mathbf{w}$ to be the weighted average of support vectors

SVC: How We Classify

Say we found the support vector and their dual variables

We set our model to

$$\mathbf{w} = \sum_{n \in \mathcal{S}} \lambda_n v_n \mathbf{x}_n$$

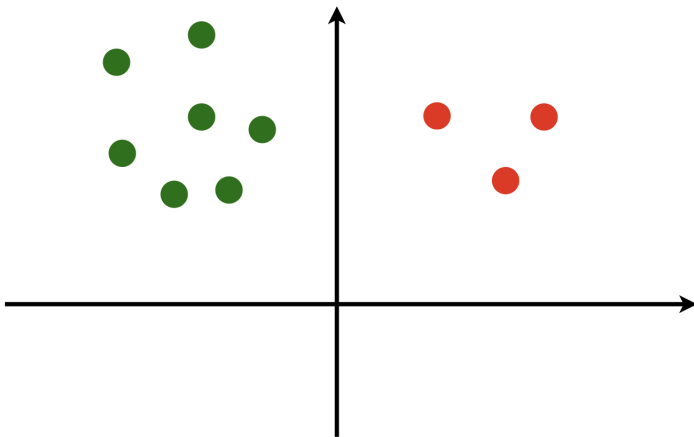
? *How do we classify a new sample $\mathbf{x}$?*

We should check the sign of

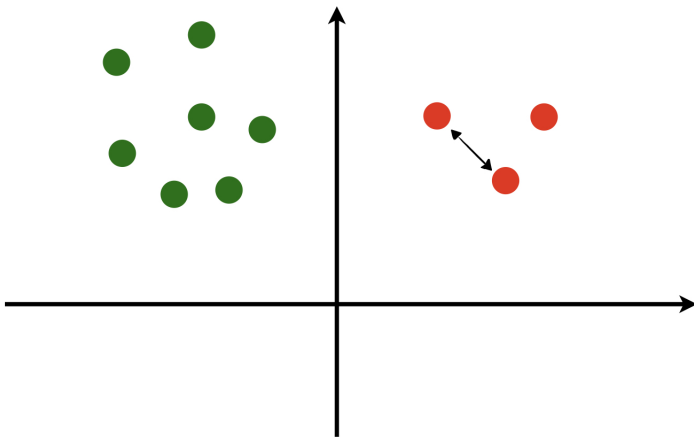
$$y = \mathbf{w}^\top \mathbf{x} = \sum_{n \in \mathcal{S}} \lambda_n v_n \mathbf{x}_n^\top \mathbf{x}$$

*We again need only to know the **cross-correlations!***

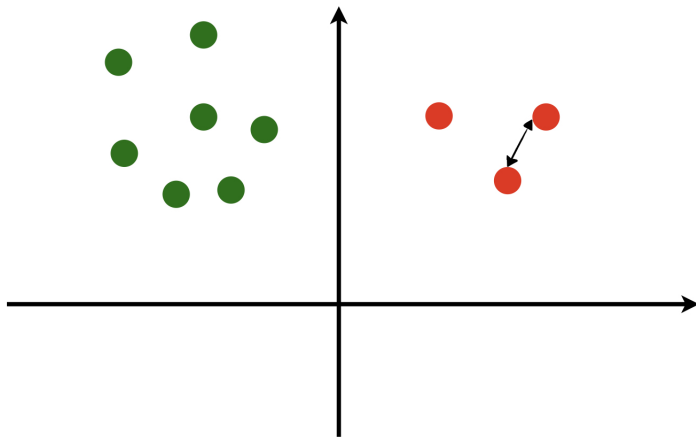
SVC Training: *Visualization*



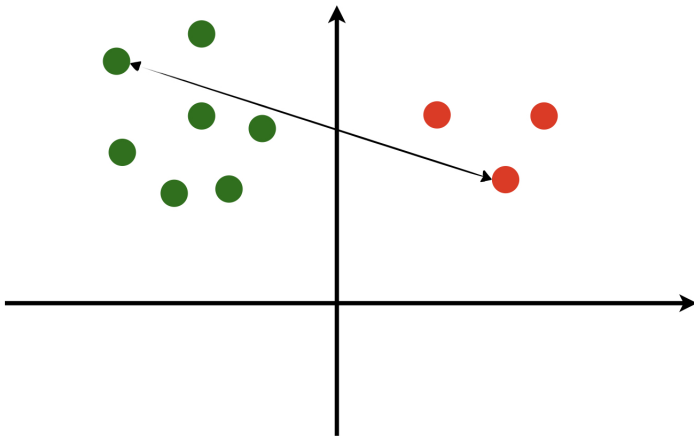
SVC Training: *Visualization*



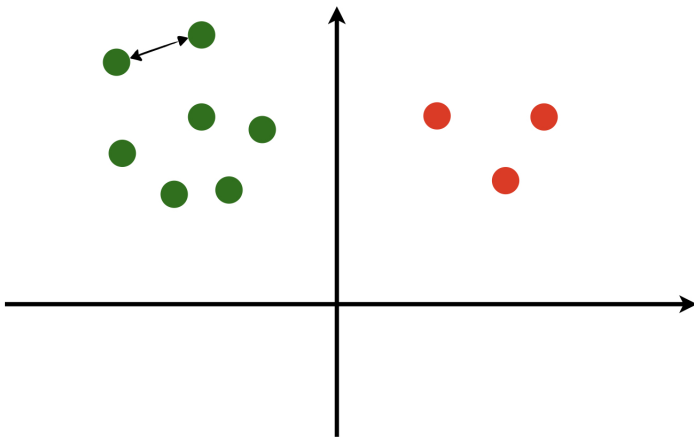
SVC Training: *Visualization*



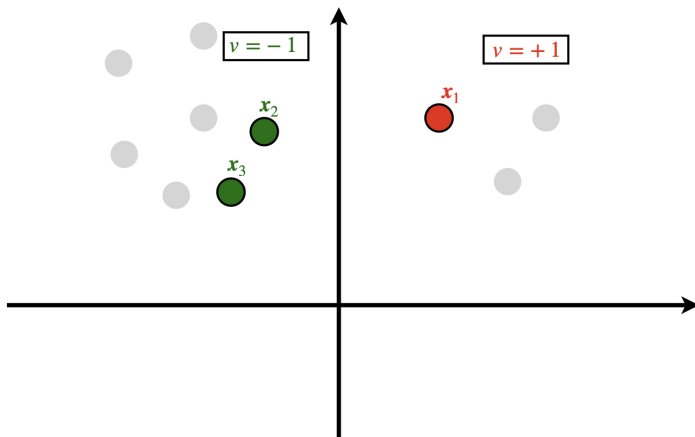
SVC Training: *Visualization*



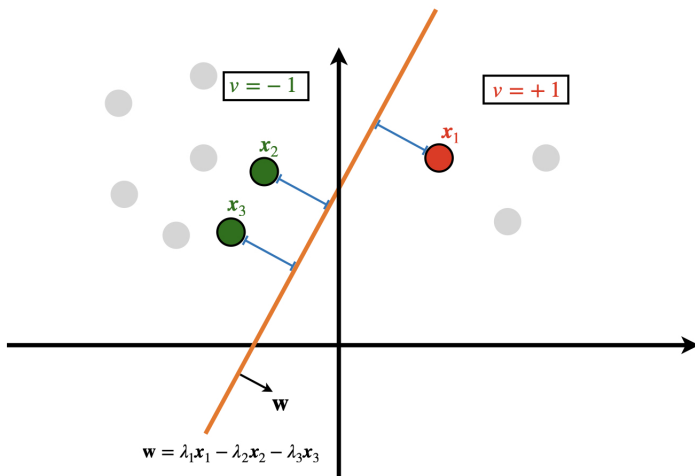
SVC Training: *Visualization*



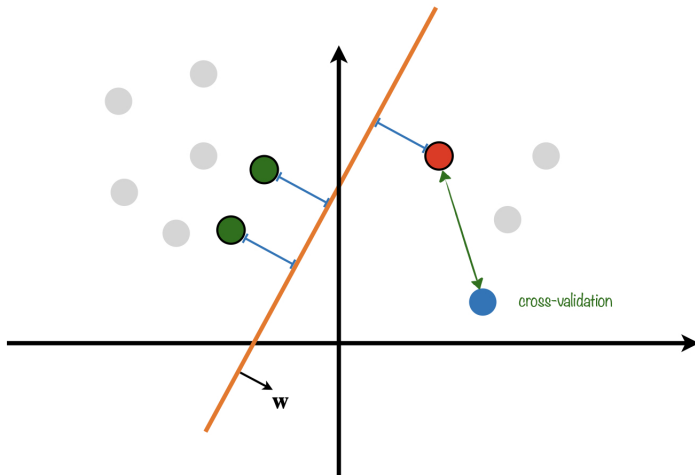
SVC Training: *Visualization*



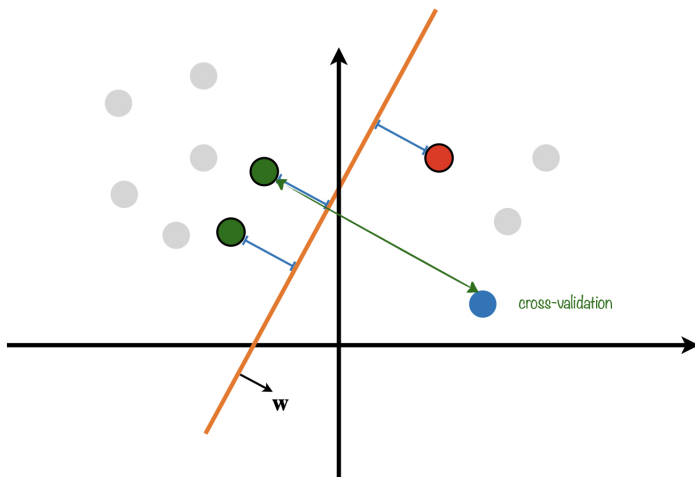
SVC Training: *Visualization*



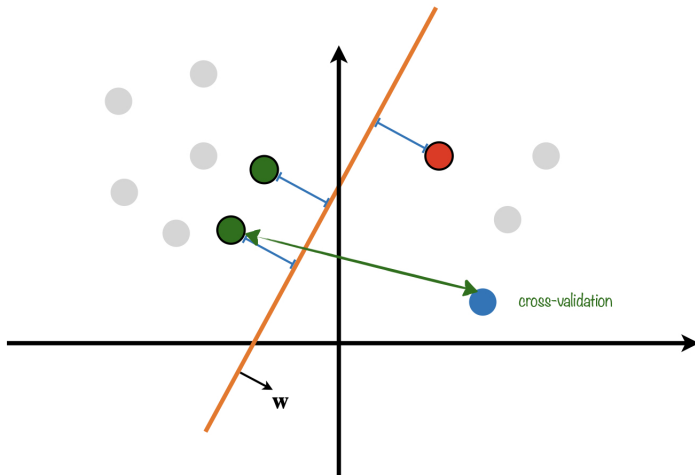
SVC Inference: *Visualization*



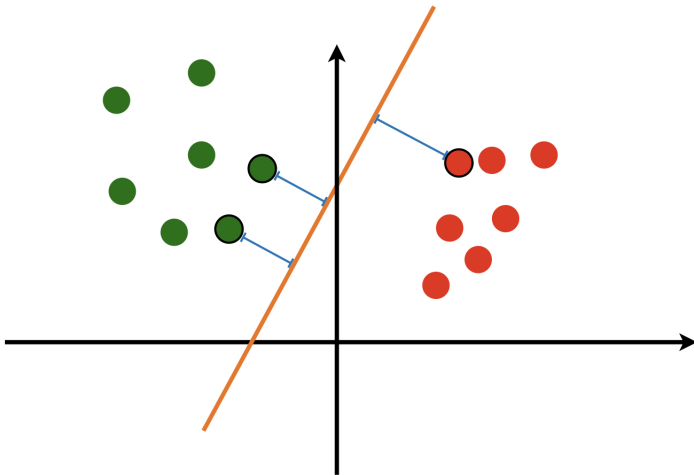
SVC Inference: *Visualization*



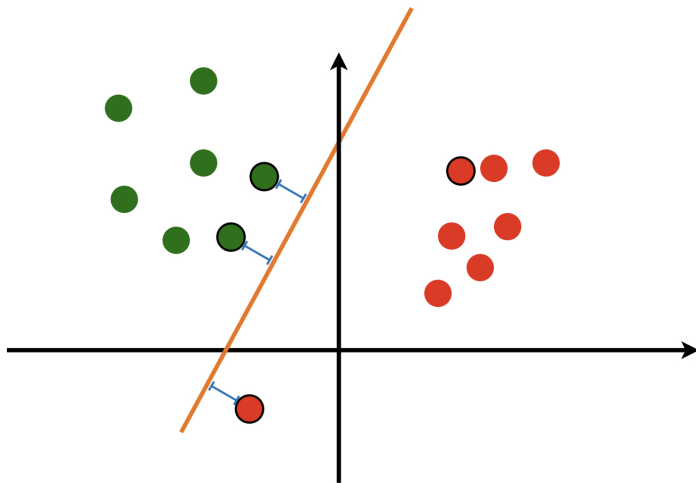
SVC Inference: *Visualization*



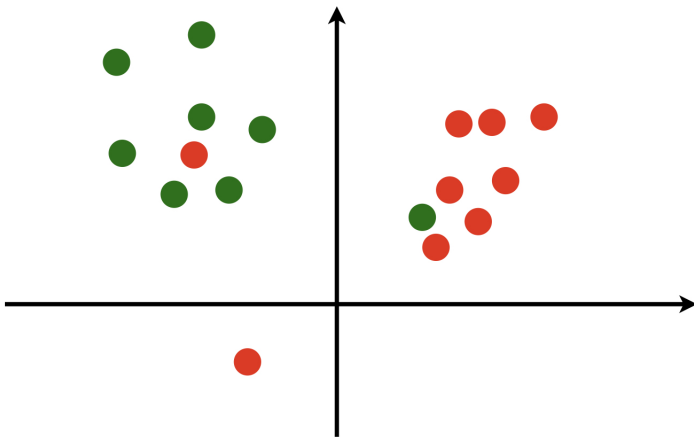
Challenges with SVC



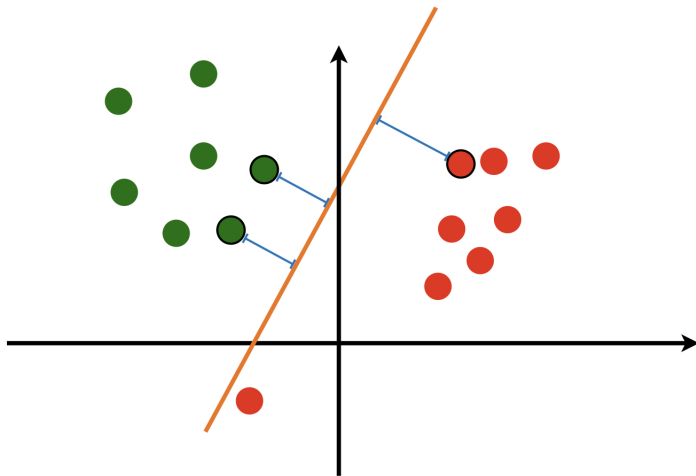
Challenges with SVC: *Outliers*



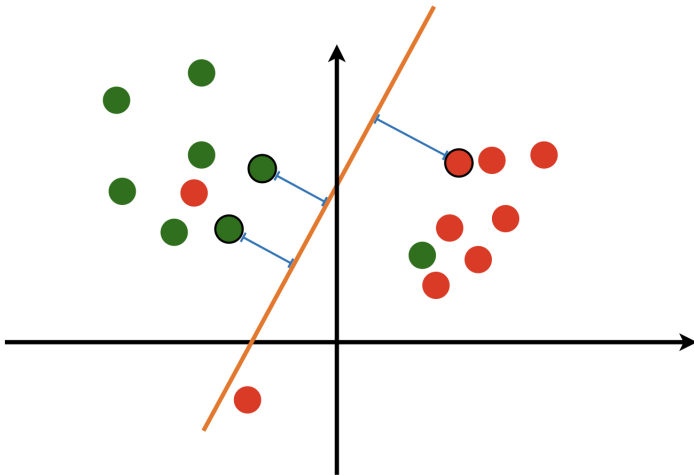
Challenges with SVC: *Nonlinear Problem*



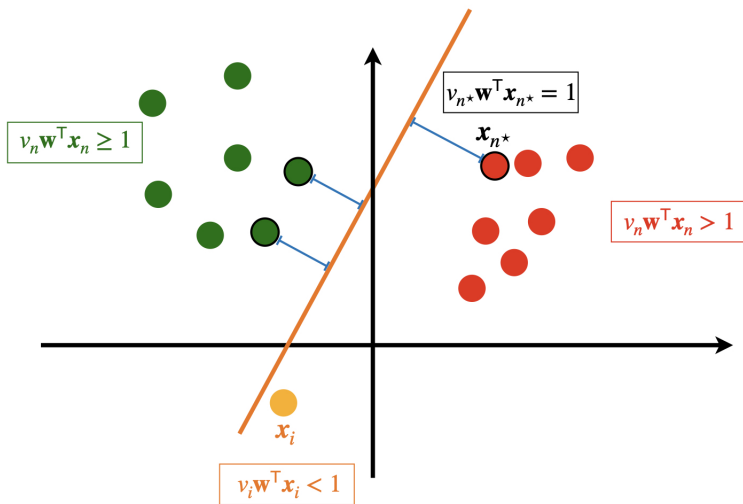
Solution: *Soft Margin*



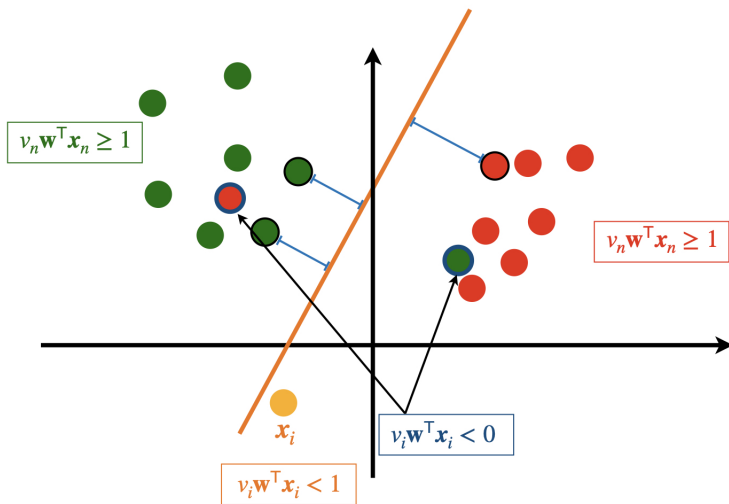
Soft Margin: *Trade-Off*



Soft Margin: *Formulation*



Soft Margin: *Formulation*



SVC with Soft Margin

Only the constraints that are impacted: so, we can write

$$\min_{\mathbf{w}} \frac{1}{2} \|\mathbf{w}\|^2 \quad \text{subject to } v_n \mathbf{w}^\top \mathbf{x}_n \geq 1 - \xi_n \text{ for all } n$$

for some $\xi_n \geq 0$

- if $\xi_n = 0 \rightsquigarrow \mathbf{x}_n$ is *on or out of margin*
- if $0 < \xi_n \leq 1 \rightsquigarrow \mathbf{x}_n$ is *in the margin*
- if $\xi_n > 1 \rightsquigarrow \mathbf{x}_n$ is *misclassified*

SVC with Soft Margin

We can let $\xi_n \geq 0$ to be tuned: we solve

$$\min_{\mathbf{w}, \xi_n \geq 0} \frac{1}{2} \|\mathbf{w}\|^2 + C \sum_{n=1}^N \xi_n \quad \text{subject to } v_n \mathbf{w}^\top \mathbf{x}_n \geq 1 - \xi_n \text{ for all } n$$

Here, C controls misclassification level

- if $C \rightarrow 0 \rightsquigarrow \xi_n$ can be large
 - ↳ We have **lots of** misclassification
- if $C \rightarrow \infty \rightsquigarrow \xi_n$ should be small
 - ↳ We have **less** misclassification

Further Read

- Bishop

- ↳ Chapter 6: *Section 6.1*

SVC Solution

- ESL

- ↳ Chapter 12: *Section 12.1 - 12.2*

SVC